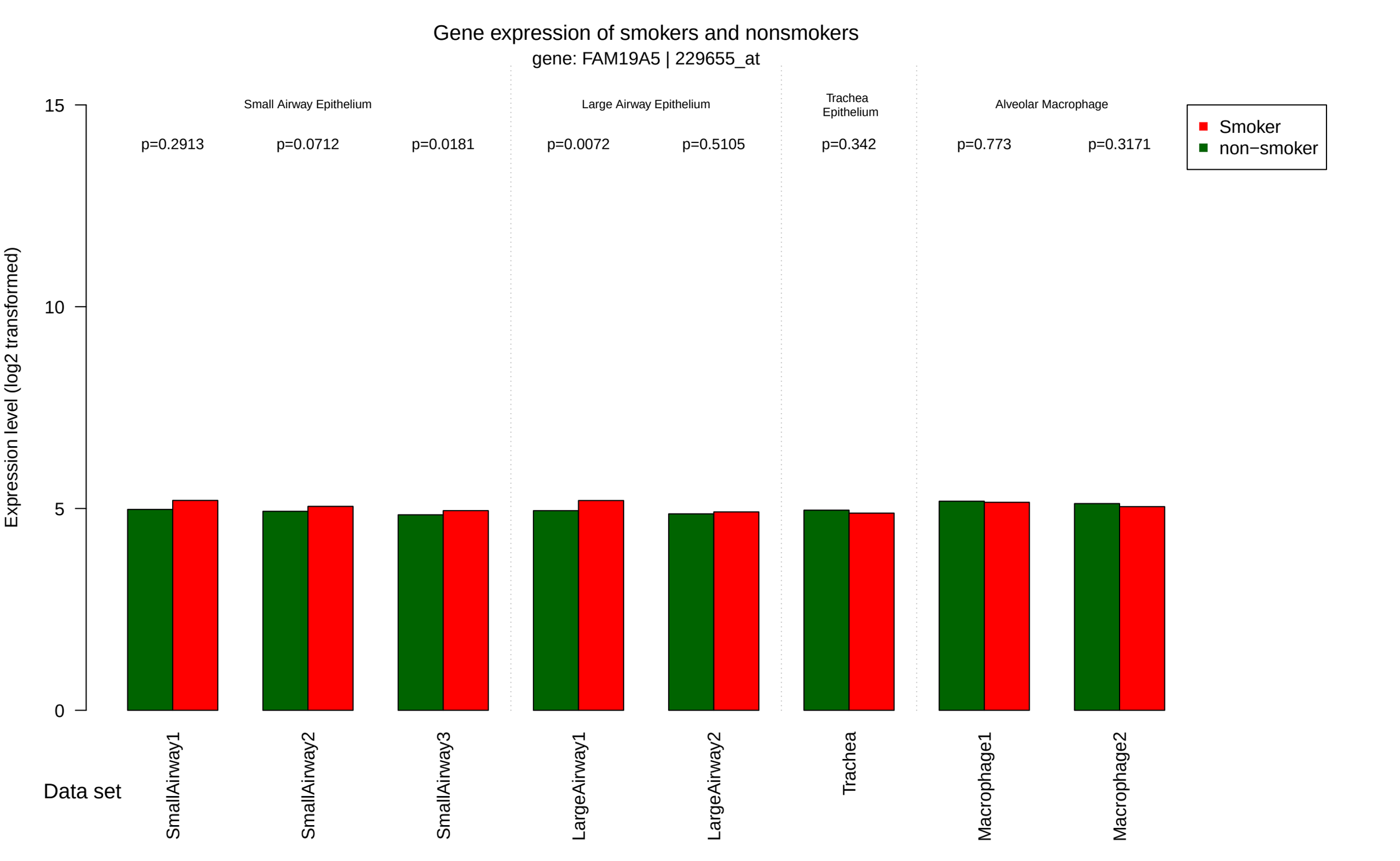
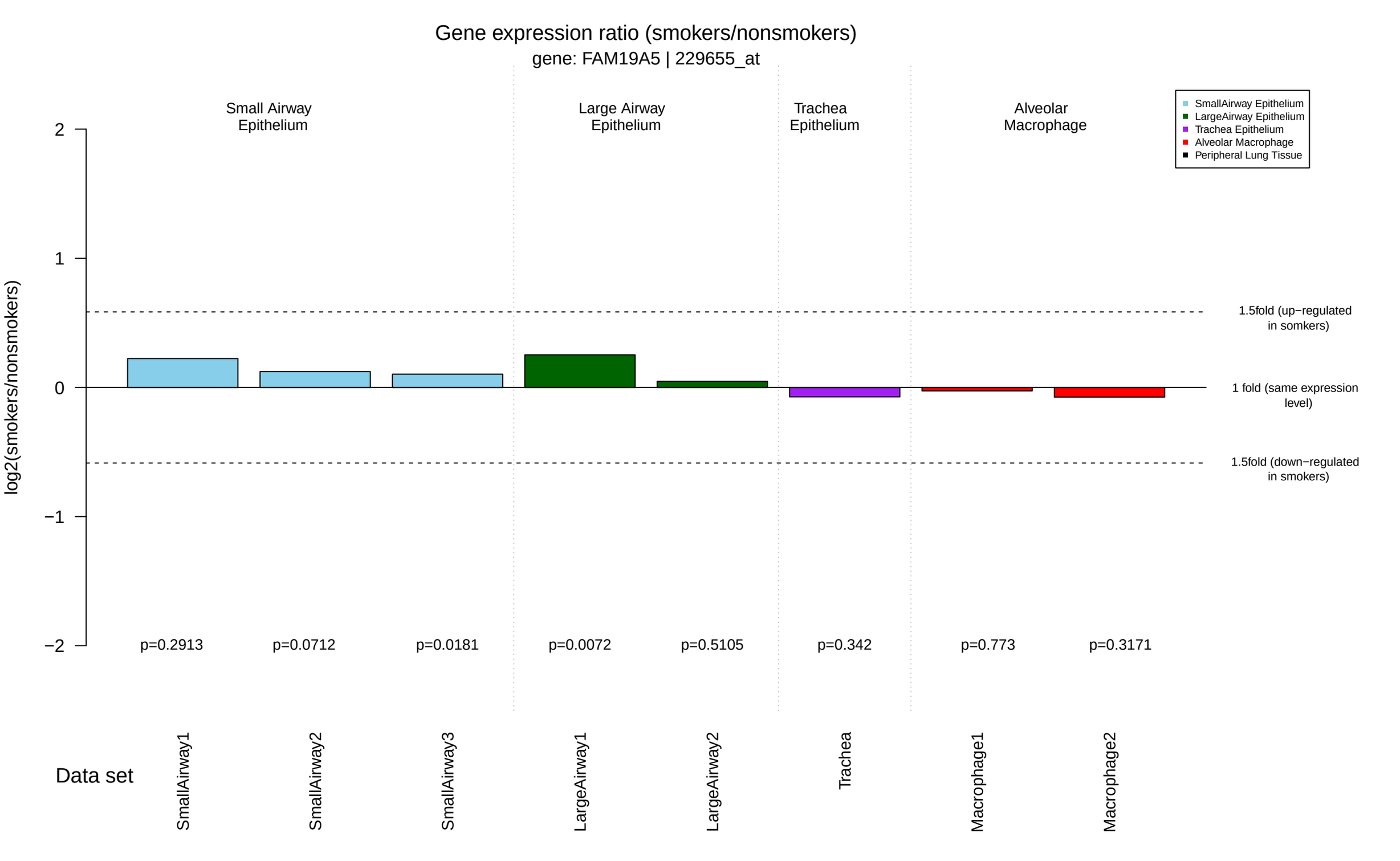
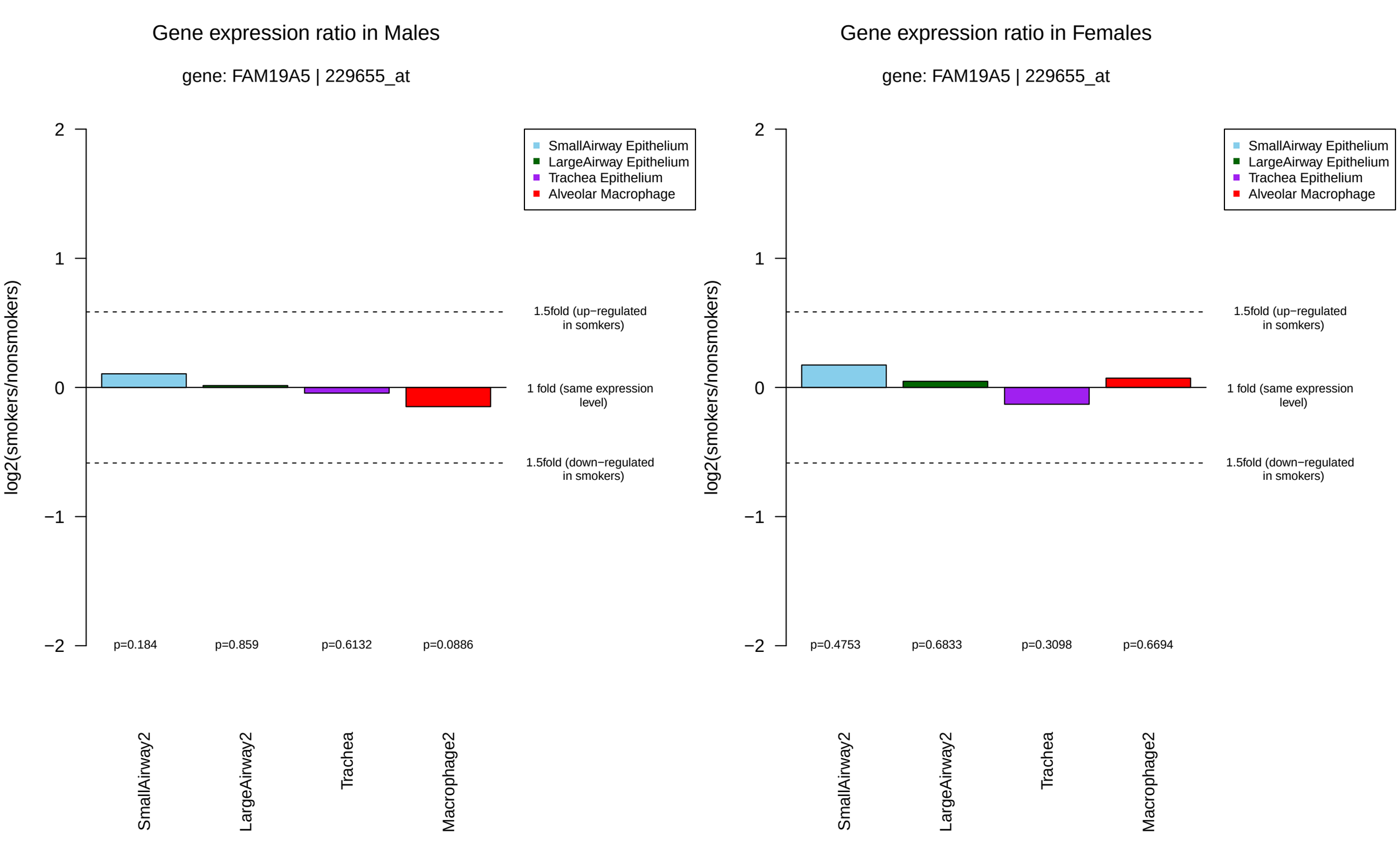
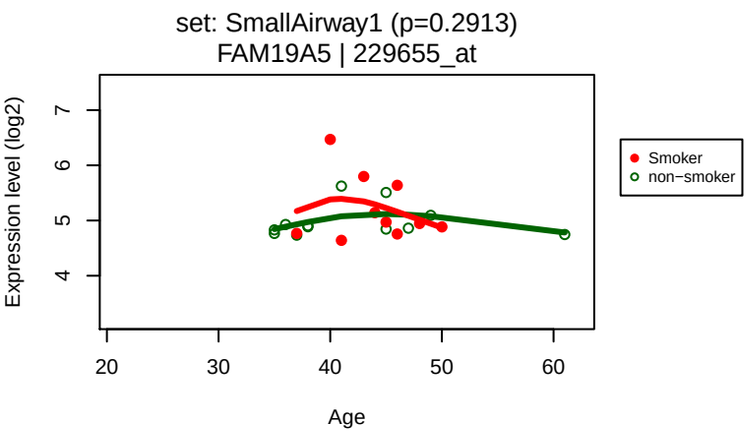
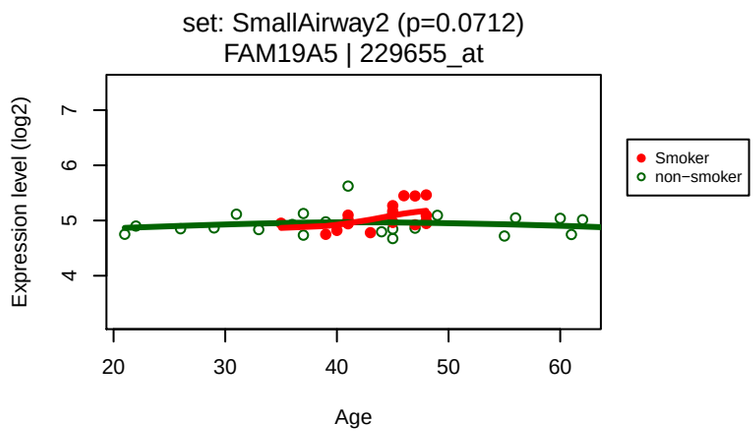
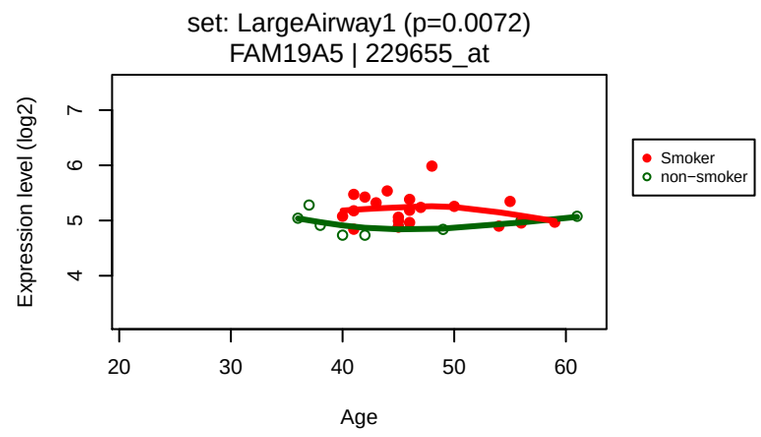

Gene expression of smokers and nonsmokers
gene: FAM19A5 | 229655_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2913
p=0.0712
p=0.0181
p=0.0072
p=0.5105
p=0.342
p=0.773
p=0.3171
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

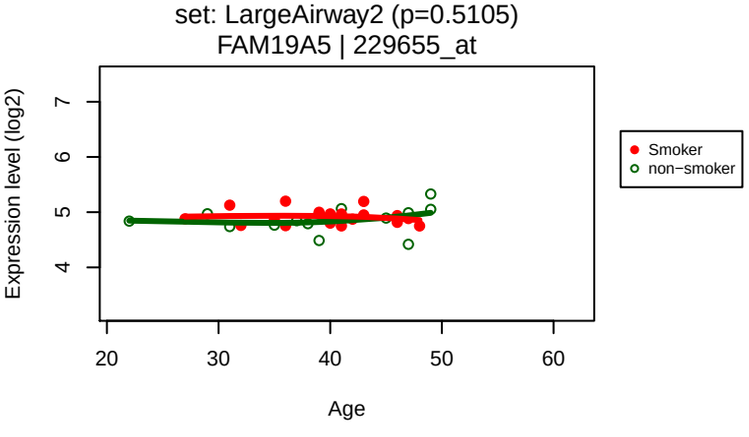

Gene expression ratio (smokers/nonsmokers)
gene: FAM19A5 | 229655_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2913
p=0.0712
p=0.0181
p=0.0072
p=0.5105
p=0.342
p=0.773
p=0.3171
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

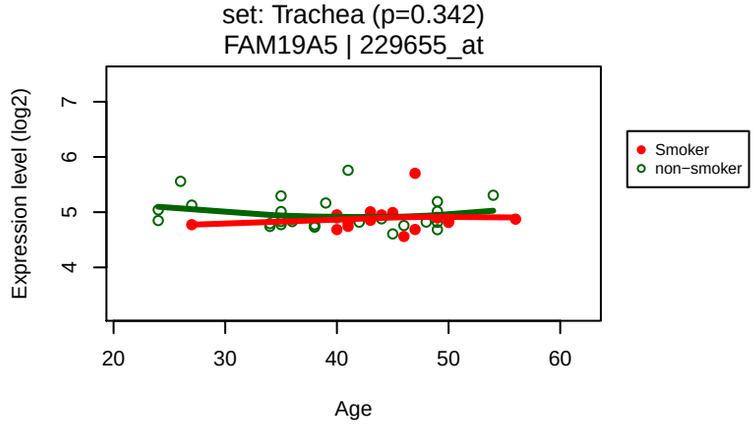

Gene expression ratio in Males
gene: FAM19A5 | 229655_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.184
p=0.859
p=0.6132
p=0.0886
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FAM19A5 | 229655_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4753
p=0.6833
p=0.3098
p=0.6694
SmallAirway2
LargeAirway2
Trachea
Macrophage2

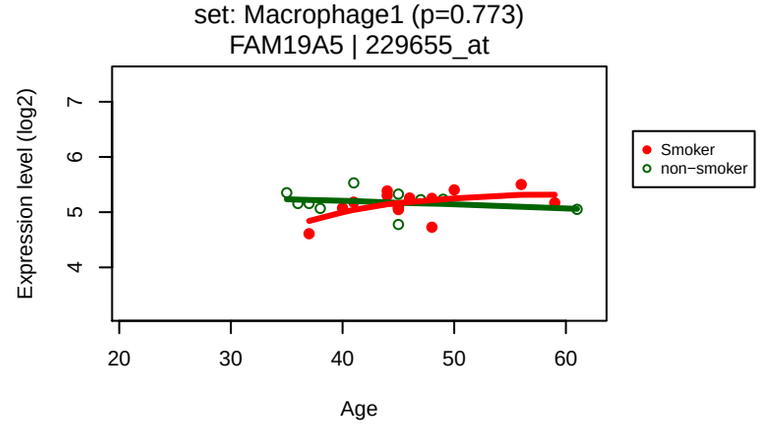

set: SmallAirway1 (p=0.2913)
FAM19A5 | 229655_at
Expression level (log2)
Age
Smoker
non−smoker

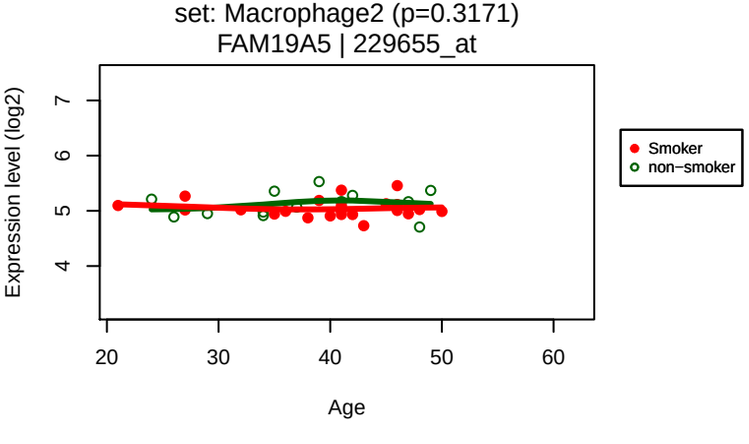

set: SmallAirway2 (p=0.0712)
FAM19A5 | 229655_at
Expression level (log2)
Age
Smoker
non−smoker

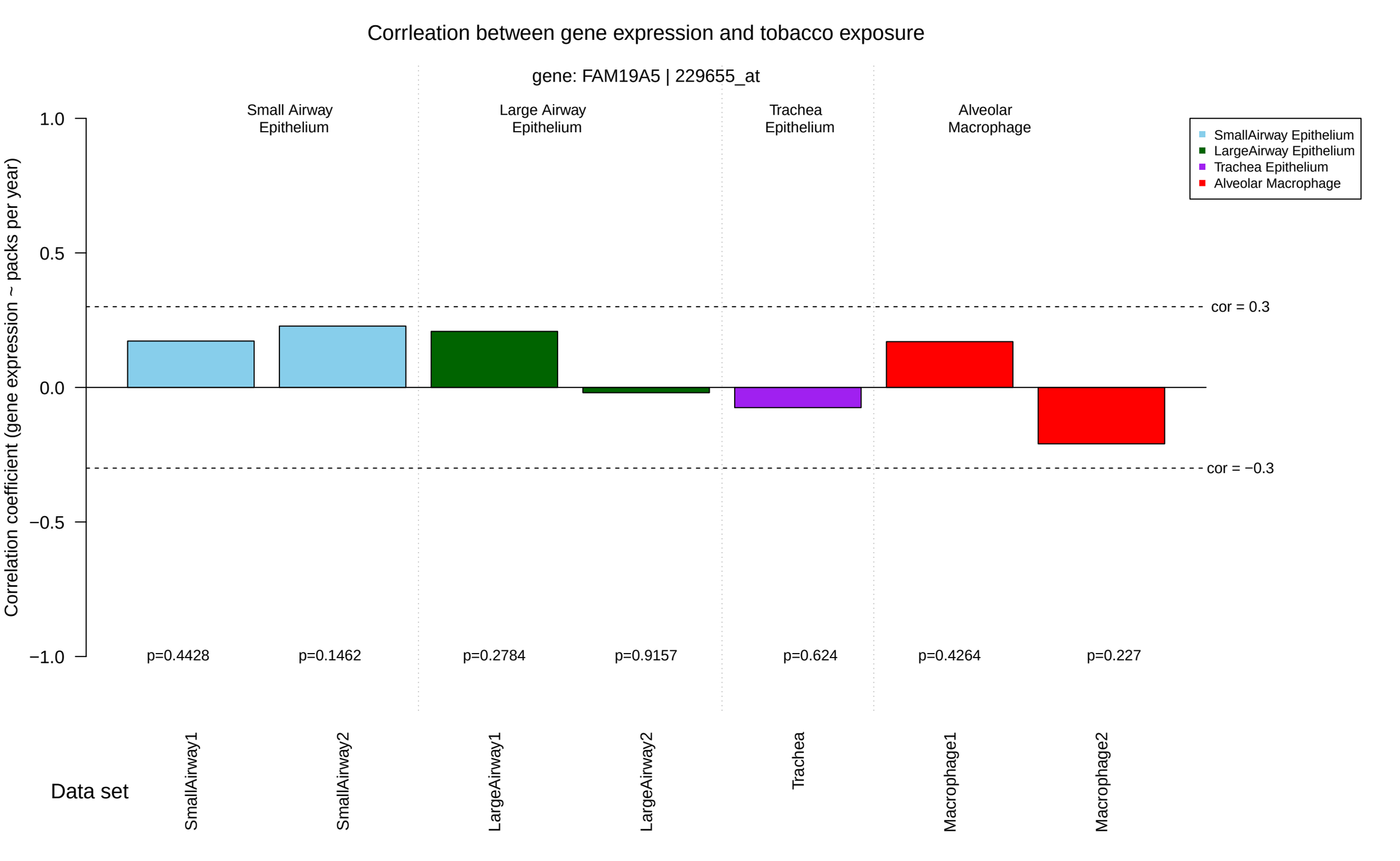

set: LargeAirway1 (p=0.0072)
FAM19A5 | 229655_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5105)
FAM19A5 | 229655_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.342)
FAM19A5 | 229655_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.773)
FAM19A5 | 229655_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3171)
FAM19A5 | 229655_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FAM19A5 | 229655_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4428
p=0.1462
p=0.2784
p=0.9157
p=0.624
p=0.4264
p=0.227
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2